

SynthoGestures: A Novel Framework for Synthetic Dynamic Hand Gesture Generation for Driving Scenarios

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ABSTRACT

Creating a diverse and comprehensive dataset of hand gestures for dynamic human-machine interfaces in the automotive domain can be challenging and time-consuming. To overcome this challenge, we propose using synthetic hand gestures generated by virtual 3D models. In this paper, we present our open-source framework that utilizes Unreal Engine to synthesize realistic static and dynamic hand gestures, offering customization options and reducing the risk of overfitting. Multiple variants, including gesture speed, performance, and hand shape, are generated to improve generalizability. In addition, we simulate different camera locations and types, such as RGB, infrared, and depth cameras, without incurring additional time, effort, or cost to obtain these cameras. Experimental results demonstrate that our proposed framework, SynthoGestures, improves gesture recognition accuracy and can replace or augment real-hand datasets. By saving time and effort in the creation of a data set, our tool accelerates the development of gesture recognition systems for automotive and non-automotive applications.

CCS CONCEPTS

- **Human-centered computing** → **Gestural input**; *User models*;
- **Computing methodologies** → **Neural networks**.

KEYWORDS

Gesture Recognition; Synthetic data; Data Augmentation; Personalization; Deep Learning

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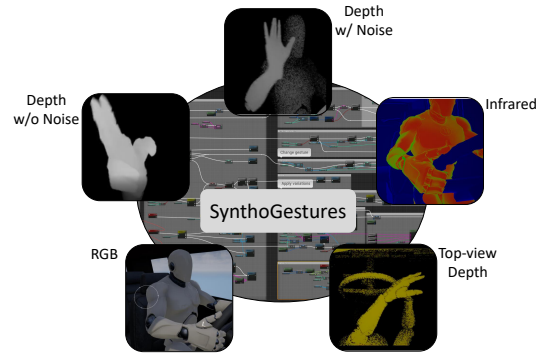


Figure 1: We present, a novel 3D synthetic hand gesture generation framework, *SynthoGestures*, that provides a cost-effective and flexible approach for creating new variations of dynamic and static hand gestures.

1 INTRODUCTION AND RELATED WORK

Hand gestures are an essential aspect of human-machine interaction, as they allow natural communication between humans and devices [1, 2, 4, 8, 15, 23]. In particular, hand gesture recognition can control various parts of the vehicle, such as audio systems, climate control, and navigation [16, 21, 22]. Hand gesture recognition involves detecting, analyzing, and interpreting hand movements to understand the intended message [3, 11, 12, 14, 20, 23]. As technology advances rapidly, hand gesture recognition techniques have evolved significantly. Despite these advances, state-of-the-art techniques rely on complex deep learning algorithms that require large amounts of data [8, 9, 15, 17]. While most existing state-of-the-art gesture recognition methods achieve high performance, they suffer from a lack of generalization due to multiple factors, such as the need for further data and dataset biases. Several approaches have explored synthesizing hand gestures to improve gesture recognition models [5–7, 10, 13, 18, 19]. However, most of the approaches focused on static, isolated hand gesture recognition with limited to no background noise. Thus, we propose *SynthoGestures*, an Unreal Engine-based framework to create dynamic hand gestures with several simulated background noise and hand variations to be used in training large gesture recognition models. Our tool can be used

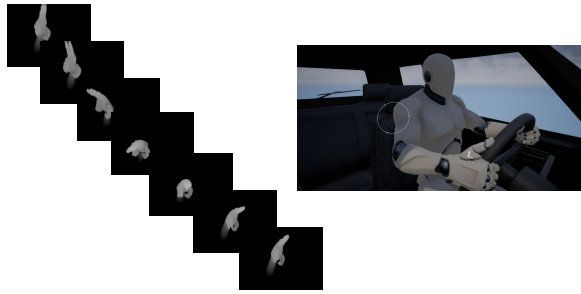


Figure 2: An example of the character model with the path of a rotation gesture (right) with the generated gesture sequence as a low frame rate depth image without noise (left).

to create synthetic data from multiple camera positions within the vehicle (i.e. multiple point of view), as well as multiple sensor types, such as RGB, infrared, and depth-based cameras, with added noise modeled from existing hardware. Additionally, we provide the first batch of synthetic hand gestures in the automotive domain and highlight its quality when compared to a real hand gesture dataset.

2 METHOD

Our framework¹ employs a systematic approach to generate dynamic hand gestures. It starts by collecting initial settings from user input or a JSON file, including recording parameters and default hand selection. Users can customize gestures by specifying camera types, locations, hand attributes, gesture speed, and lighting conditions. The framework automates the generation process by iterating through gesture-specific settings, resulting in a comprehensive dataset with various performance variations. More specifically, users can customize camera setups including camera types (e.g., RGB, depth, and infrared camera) and their associated parameters. Once the desired camera angle is selected, users can create their own gestures or select from existing range of gestures, such as horizontal swipe, vertical swipe, rotate right, rotate left, and pointing gestures. Several additional parameters can be chosen for the gesture, including *Gesture Name*, *Number Of Variations*, *Use of Left Hand*, *Gesture Speed*, *Pre-gesture Speed*, and *Post-gesture Speed* as well as other several aspects of gesture performance and the physique of the hand. Figure 2 shows an example of a rotation gesture generated with our framework.

3 RESULTS AND DISCUSSION

To assess the effectiveness of the gesture generation system, we employ the methodologies of Tsai et al. [19] and Ibrahim and Kashef [7] in combining synthetic and real data to improve the accuracy of recognition or maintain performance with limited real data points. We use NVIDIA’s Dynamic Hand Gesture Dataset [14] and the state-of-the-art gesture recognition model developed by Köpüklü et al. [11, 12] as real data. We generate their counterpart using our framework with multiple variations and a depth camera simulated setting using our framework. Figure 3 presents the recognition accuracy of these experiments. It is evident that a pretraining strategy

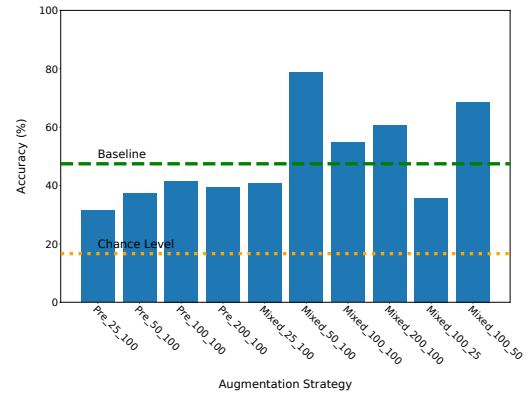


Figure 3: Gesture recognition accuracy for different combination of synthetic and real hand gestures. The prefix “Pre” are the models pre-trained with the synthetic data, while “Mixed” are models trained from scratch with the synthetic and real data combined. The first number is the percentage of synthetically generated hand gestures while the second number is for the real data. The total number of gestures in the real data is 33 gestures per class (a 200% value in the synthetic gesture indicates twice more generated data). “Baseline” is the model trained on real data only.

did not improve accuracy compared to baseline, indicating that the use of synthetic data alone is not equivalent to using real data. However, training the model from scratch with a combination of synthetic and real data demonstrated superior performance, particularly when a substantial amount of synthetic data was incorporated. These experiments highlight that models incorporating synthetic with real data outperform those trained solely on real data. This capability saves significant time and effort in the creation of datasets and accelerates the development of gesture recognition systems. Although these results have demonstrated the effectiveness of SynthoGestures in improving gesture recognition performance, it is essential to acknowledge certain limitations and potential avenues for future work. First, the current framework focuses primarily on the visual aspect of hand gestures and may not capture other sensory cues, such as tactile or proprioceptive feedback. Exploring ways to incorporate these additional modalities could further enhance the authenticity and realism of synthesized gestures. Moreover, although SynthoGestures provides a wide range of variations in terms of gesture speed, performance, and hand shape, there is still room to explore more complex and nuanced variations that can capture the intricacies of natural hand movements. Thus, our framework can facilitate future work, advance the field of static and dynamic hand gesture recognition, and pave the way for more sophisticated and accurate human-computer interaction systems in various domains, including automotive interfaces, virtual reality, and augmented reality applications.

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¹<https://github.com/amrgomaaelhady/SynthoGestures>

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